

Designed a Truss Bridge in SolidWorks based on the Baltimore concept, with a clear objective — minimum weight and maximum load-bearing capacity. The final model withstands up to 100 kg without failure.

To achieve a lightweight structure, the entire bridge was built using the Weldments feature in SolidWorks.

- **Structural Planning** — Sketched one side of the bridge on the Front Plane and mirrored it to form the complete structure, maintaining symmetry throughout.
- **Chord Construction** — Used the Top Plane to construct the upper chord, then mirrored it to complete the lower chord and lateral sides.
- **Weldment Members** — Applied structural members using Trim features across multiple planes to complete the full prototype.

Skills Applied: SolidWorks Weldments · Structural Member Profiles · Mirroring · Multi-Plane Sketching · Structural Design